

Calculus 2

Sequences, Series, Power Series, Polar / Parametric Curves, ODEs

mathcheatsheet

Abstract

A working reference for the standard second-semester calculus material: sequences and their limits; infinite series and the full battery of convergence tests; power series, radius of convergence, and Taylor / Maclaurin expansions; parametric and polar curves with their calculus (slopes, arc length, area); and an introduction to first-order ordinary differential equations. Each section pairs definitions and theorems with at least one worked example and a short list of common pitfalls.

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1 Sequences

Definition 1.1 (Sequence). A sequence is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, written $\{a_n\}_{n \geq 1}$ or (a_n) . We say $a_n \rightarrow L$ (or $\lim_{n \rightarrow \infty} a_n = L$) if for every $\varepsilon > 0$ there exists N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

1.1 Limit laws and the function-limit bridge

All algebraic limit laws (sum, product, quotient with nonzero denominator) carry over from function limits. The most useful bridge:

Theorem 1.2. *If $f : [1, \infty) \rightarrow \mathbb{R}$ is such that $a_n = f(n)$ and $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} a_n = L$.*

So you can apply L'Hôpital's rule to $f(x)$ to evaluate $\lim_{n \rightarrow \infty} a_n$.

1.2 Monotone convergence

Theorem 1.3 (Monotone Convergence). *A monotone sequence converges iff it is bounded.*

1.3 Standard limits

Memorize

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \ (p > 0), \quad \lim_{n \rightarrow \infty} r^n = 0 \text{ if } |r| < 1, \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1, \\ \lim_{n \rightarrow \infty} \sqrt[n]{c} = 1 \ (c > 0), \quad \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0. \end{aligned}$$

Example 1.4. $a_n = \frac{n^2 + 3}{2n^2 - n} \rightarrow \frac{1}{2}$ (divide top and bottom by n^2).

Example 1.5 (Squeeze). $0 \leq \frac{\sin n}{n} \leq \frac{1}{n} \Rightarrow \frac{\sin n}{n} \rightarrow 0$.

Pitfalls

- Convergence of $\{a_n\}$ has nothing to do with convergence of the *series* $\sum a_n$ — that is a different question (next section).
- $a_n \rightarrow 0$ does not imply $\sum a_n$ converges (e.g. harmonic series).

2 Infinite Series

Definition 2.1 (Series and partial sums). Given $\{a_n\}$, the N -th partial sum is $S_N = \sum_{n=1}^N a_n$. The series $\sum_{n=1}^{\infty} a_n$ *converges to* S if $S_N \rightarrow S$; otherwise it diverges.

2.1 Two foundational series

Geometric series

For $|r| < 1$,

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

Diverges for $|r| \geq 1$. The N -th partial sum is $S_N = a \frac{1-r^{N+1}}{1-r}$.

p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \iff p > 1.$$

In particular the harmonic series $\sum 1/n$ diverges; $\sum 1/n^2 = \pi^2/6$ converges.

2.2 Telescoping series

If $a_n = b_n - b_{n+1}$, then $S_N = b_1 - b_{N+1}$ and the series converges iff b_n converges.

Example 2.2. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1$.

2.3 Algebraic properties

If $\sum a_n = A$ and $\sum b_n = B$ converge, then for any constants α, β

$$\sum (\alpha a_n + \beta b_n) = \alpha A + \beta B.$$

Convergence of $\sum a_n$ is unaffected by changing finitely many terms (the value changes, not the convergence).

2.4 Divergence test (the easy “no”)

Theorem 2.3 (n -th term test). If $\sum a_n$ converges, then $a_n \rightarrow 0$.

Contrapositive (what you actually use): if $a_n \not\rightarrow 0$, then $\sum a_n$ diverges. This is the first thing to check on any series.

Pitfalls

“ $a_n \rightarrow 0$ ” does *not* imply convergence. The harmonic series $\sum 1/n$ satisfies $1/n \rightarrow 0$ yet diverges. The n -th term test is one-directional.

3 Convergence Tests

A decision tree for positive-term series ($a_n \geq 0$) and one extra tool for sign-changing series.

3.1 Integral test

Theorem 3.1. *If f is positive, continuous, and decreasing on $[1, \infty)$ with $a_n = f(n)$, then $\sum a_n$ and $\int_1^\infty f(x) dx$ either both converge or both diverge.*

3.2 Comparison tests

Theorem 3.2 (Direct Comparison). *Suppose $0 \leq a_n \leq b_n$ for all $n \geq N$.*

- *If $\sum b_n$ converges, so does $\sum a_n$.*
- *If $\sum a_n$ diverges, so does $\sum b_n$.*

Theorem 3.3 (Limit Comparison). *If $a_n, b_n > 0$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ with $0 < c < \infty$, then $\sum a_n$ and $\sum b_n$ behave alike (both converge or both diverge).*

3.3 Ratio and root tests

Theorem 3.4 (Ratio Test). *Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$. Then*

$\sum a_n$ *converges absolutely if $L < 1$, diverges if $L > 1$, test inconclusive if $L = 1$.*

Theorem 3.5 (Root Test). *Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Same conclusions as the ratio test ($L < 1$ converges absolutely, $L > 1$ diverges, $L = 1$ inconclusive).*

Use the ratio test when a_n involves factorials or exponentials; the root test when a_n has a clean n -th power.

3.4 Alternating Series Test (Leibniz)

Theorem 3.6. *If $b_n > 0$, b_n is eventually decreasing, and $b_n \rightarrow 0$, then*

$$\sum_{n=1}^{\infty} (-1)^n b_n \quad \text{converges.}$$

Moreover the truncation error $|S - S_N| \leq b_{N+1}$.

3.5 Absolute vs. conditional convergence

- $\sum a_n$ converges **absolutely** if $\sum |a_n|$ converges. Absolute convergence $\Rightarrow$ convergence.
- It converges **conditionally** if it converges but not absolutely (e.g. $\sum (-1)^n/n$).
- Riemann rearrangement: a conditionally convergent real series can be rearranged to sum to *any* real number, or to diverge. Absolutely convergent series cannot.

3.6 Strategy

Picking a test

1. **n -th term test** — does $a_n \rightarrow 0$? If not, diverges.
2. **Geometric / p -series** — can you recognize the form?
3. **Alternating** signs $\Rightarrow$ try Leibniz.

4. **Factorials / exponentials** $\Rightarrow$ ratio test.
5. **Powers of n** (like $a_n = b_n^n$) $\Rightarrow$ root test.
6. **Rational in n / polynomials** $\Rightarrow$ limit comparison with a p -series.
7. $a_n = f(n)$ with f easy to integrate $\Rightarrow$ integral test.

Example 3.7 (Ratio test). $\sum \frac{n!}{n^n} : \left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n^n}{(n+1)^n} = \frac{1}{(1+1/n)^n} \rightarrow 1/e < 1$. Converges.

4 Power Series

Definition 4.1. A power series centered at a is

$$\sum_{n=0}^{\infty} c_n(x-a)^n.$$

Its set of x where the series converges is an interval centered at a ; the half-width is the *radius of convergence* $R \in [0, \infty]$.

4.1 Finding the radius of convergence

Apply the ratio (or root) test to the variable- x series:

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \quad \text{or} \quad \frac{1}{R} = \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}.$$

(Use $R = \infty$ if the limit is 0, $R = 0$ if it is ∞ .) The series converges absolutely on $(a-R, a+R)$ and diverges for $|x-a| > R$. Behavior at the endpoints $x = a \pm R$ must be checked separately.

Example 4.2. $\sum_{n=0}^{\infty} x^n/n!$: ratio $= |x|/(n+1) \rightarrow 0$ for all x . So $R = \infty$ and the series is e^x .

Example 4.3. $\sum_{n=1}^{\infty} x^n/n$: ratio $= |x| \cdot n/(n+1) \rightarrow |x|$, so $R = 1$. At $x = 1$ harmonic (diverges); at $x = -1$ alternating harmonic (converges conditionally). Interval $[-1, 1)$.

4.2 Algebra and calculus of power series

Inside the open interval of convergence, a power series defines an infinitely differentiable function; you can differentiate and integrate term-by-term, and the result has the same radius of convergence (endpoints may change):

$$f(x) = \sum c_n(x-a)^n \Rightarrow f'(x) = \sum n c_n(x-a)^{n-1}, \quad \int f dx = C + \sum \frac{c_n}{n+1}(x-a)^{n+1}.$$

Example 4.4 (Trick). $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$. Differentiate: $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$. Integrate: $-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$.

4.3 Multiplying power series

$$\left(\sum a_n x^n \right) \left(\sum b_n x^n \right) = \sum_{n=0}^{\infty} c_n x^n, \quad c_n = \sum_{k=0}^n a_k b_{n-k} \text{ (Cauchy product).}$$

Pitfalls

- Endpoints must be tested by hand — the ratio test never decides them.
- Term-by-term differentiation works only on the *open* interval $(a - R, a + R)$.
- Substitution: if you know a series for $f(x)$, then $f(g(x))$ may be obtained by substitution only when $|g(x)| < R$.

5 Taylor and Maclaurin Series

Definition 5.1 (Taylor series). If f has derivatives of all orders at a , its Taylor series at a is

$$T_f(x; a) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

A *Maclaurin series* is the special case $a = 0$.

5.1 Taylor's theorem with remainder

For f that is $(n + 1)$ -times differentiable on an interval containing a and x :

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + R_n(x),$$

with the Lagrange form of the remainder

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1}, \quad \xi \text{ between } a \text{ and } x.$$

If $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then f equals its Taylor series on that interval.

5.2 Essential Maclaurin series

Memorize — all centered at 0

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} && \text{all } x \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} && \text{all } x \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} && \text{all } x \\ \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n && |x| < 1 \\ \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} && -1 < x \leq 1 \\ \arctan x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} && |x| \leq 1 \\ (1+x)^\alpha &= \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n && |x| < 1 \end{aligned}$$

where $\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$ (binomial series).

5.3 Using known series

Almost every Taylor problem on an exam is solved by manipulating the table above: substitute, differentiate, integrate, or multiply.

Example 5.2 (Substitution). $e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}$.

Example 5.3 (Indefinite integration via series). $\int e^{-x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)}$.

Example 5.4 (Limits via series). $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$: $\sin x = x - \frac{x^3}{6} + O(x^5)$, so numerator $= -x^3/6 + O(x^5)$, ratio $\rightarrow -1/6$.

5.4 Error estimation

For an alternating Taylor series with terms decreasing in magnitude, the error after N terms is at most the first omitted term (alternating series remainder). Otherwise use the Lagrange remainder bound:

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}, \quad M = \max_{|\xi-a| \leq |x-a|} |f^{(n+1)}(\xi)|.$$

Pitfalls

A function can have a Taylor series at a that converges everywhere but does *not* equal the function (e.g. $f(x) = e^{-1/x^2}$ at 0). Equality requires $R_n(x) \rightarrow 0$.

6 Parametric Curves

A parametric curve is $(x(t), y(t))$ for t in some interval I . Same trace can arise from many parametrizations.

6.1 Slope and second derivative

Assuming $x'(t) \neq 0$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)}, \quad \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{1}{x'(t)} \frac{d}{dt} \left(\frac{y'(t)}{x'(t)} \right).$$

6.2 Arc length

For a smooth curve with x', y' continuous on $[\alpha, \beta]$:

$$L = \int_{\alpha}^{\beta} \sqrt{x'(t)^2 + y'(t)^2} dt.$$

6.3 Area under a parametric curve

If $y \geq 0$ and $x(t)$ is monotone on $[\alpha, \beta]$:

$$A = \int_{\alpha}^{\beta} y(t) x'(t) dt.$$

(Mind the sign of $x'(t)$ and the direction of traversal.)

6.4 Surface of revolution (about the x -axis)

$$S = 2\pi \int_{\alpha}^{\beta} y(t) \sqrt{x'(t)^2 + y'(t)^2} dt.$$

Example 6.1 (Cycloid). $x = r(t - \sin t)$, $y = r(1 - \cos t)$, $t \in [0, 2\pi]$. Then $x' = r(1 - \cos t)$, $y' = r \sin t$, and

$$L = \int_0^{2\pi} \sqrt{r^2(1 - \cos t)^2 + r^2 \sin^2 t} dt = r \int_0^{2\pi} \sqrt{2 - 2 \cos t} dt = 8r.$$

Pitfalls

A parametric curve can retrace itself or cross itself — the integrals above measure the curve *as traversed*, multiplicities included.

7 Polar Coordinates

7.1 Definition

(r, θ) with $x = r \cos \theta$, $y = r \sin \theta$. Conversion back: $r^2 = x^2 + y^2$, $\tan \theta = y/x$ (with quadrant chosen by signs of x, y).

7.2 Slope of a polar curve $r = f(\theta)$

Treat θ as the parameter: $x = f(\theta) \cos \theta$, $y = f(\theta) \sin \theta$.

$$\frac{dy}{dx} = \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta}.$$

7.3 Area in polar coordinates

The area swept by $r = f(\theta) \geq 0$ for $\theta \in [\alpha, \beta]$ (no self-overlap) is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} f(\theta)^2 d\theta.$$

For the region between two polar curves $r = R(\theta) \geq r = r(\theta) \geq 0$:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} (R(\theta)^2 - r(\theta)^2) d\theta.$$

7.4 Polar arc length

$$L = \int_{\alpha}^{\beta} \sqrt{f(\theta)^2 + f'(\theta)^2} d\theta.$$

7.5 Useful curves

Curve	$r(\theta)$
Circle, radius a , centered at origin	$r = a$
Circle through origin, radius $a/2$	$r = a \cos \theta$ or $a \sin \theta$
Cardioid	$r = a(1 + \cos \theta)$
Limaçon	$r = a + b \cos \theta$
Rose, n petals if odd, $2n$ if even	$r = a \cos(n\theta)$
Lemniscate	$r^2 = a^2 \cos(2\theta)$
Spiral of Archimedes	$r = a\theta$

Example 7.1 (Cardioid area). $r = 1 + \cos \theta$ on $[0, 2\pi]$:

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2}\right) d\theta = \frac{3\pi}{2}.$$

Pitfalls

- (r, θ) is not unique: $(r, \theta) = (-r, \theta + \pi)$ and $(r, \theta + 2\pi k)$ give the same point.
- Setting $r = f(\theta) = 0$ gives the values of θ where the curve passes through the pole — useful for finding limits of integration.
- To find intersections of two polar curves, also check whether both pass through the pole; the system $r_1 = r_2$ may miss intersections at the origin.

8 First-Order ODEs

A first-order ODE has the form $y' = F(x, y)$. The general solution contains one arbitrary constant; an initial condition $y(x_0) = y_0$ selects a particular solution.

8.1 Separable equations

$$\frac{dy}{dx} = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x) dx + C.$$

Example 8.1 (Exponential growth/decay). $\frac{dy}{dx} = ky \Rightarrow y = Ce^{kx}$.

8.2 First-order linear: integrating factor

$$y' + P(x)y = Q(x).$$

Let $\mu(x) = \exp(\int P(x) dx)$. Multiplying through gives $(\mu y)' = \mu Q$, so

$$y(x) = \frac{1}{\mu(x)} \left(\int \mu(x) Q(x) dx + C \right).$$

Example 8.2. $y' + 2xy = x \Rightarrow \mu = e^{x^2} \Rightarrow (e^{x^2}y)' = xe^{x^2} \Rightarrow y = \frac{1}{2} + Ce^{-x^2}$.

8.3 Exact equations

$M(x, y) dx + N(x, y) dy = 0$ is *exact* iff $\partial_y M = \partial_x N$ on a simply connected region. Then there is $F(x, y)$ with $\partial_x F = M$ and $\partial_y F = N$; solutions are $F(x, y) = C$.

8.4 Direction fields, equilibria, and stability

For $y' = f(y)$ (autonomous):

- Equilibria: zeros of f .
- An equilibrium y_* is asymptotically stable if $f'(y_*) < 0$, unstable if $f'(y_*) > 0$.

8.5 Numerical: Euler's method

For $y' = f(x, y)$, $y(x_0) = y_0$, and step h :

$$x_{n+1} = x_n + h, \quad y_{n+1} = y_n + h f(x_n, y_n).$$

Local truncation error $O(h^2)$, global error $O(h)$. Improved Euler and Runge–Kutta methods trade compute for higher order.

8.6 Modeling cheatsheet

Recipes

- **Exponential growth/decay:** $y' = ky$, $y = y_0 e^{kt}$.
- **Logistic:** $y' = ky(1 - y/M)$, $y = \frac{M}{1 + Ae^{-kt}}$.
- **Newton's law of cooling:** $T' = -k(T - T_\infty)$, $T = T_\infty + (T_0 - T_\infty)e^{-kt}$.
- **Mixing tank with constant flow:** $V' = \text{rate in} - \text{rate out}$ leads to a linear ODE in concentration.

Pitfalls

- Separating variables tacitly assumes $h(y) \neq 0$; equilibria $h(y_*) = 0$ are extra solutions you must add by hand.
- For linear ODEs the $+C$ on $\int P dx$ disappears into μ (any antiderivative works).